Lecture 7: X-ray Propagation and the Radon Transformation

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Beer-Lambert law
- Tomography
- Radon transform
- Central slice theorem (Fourier slice theorem)

Propagation of X-rays

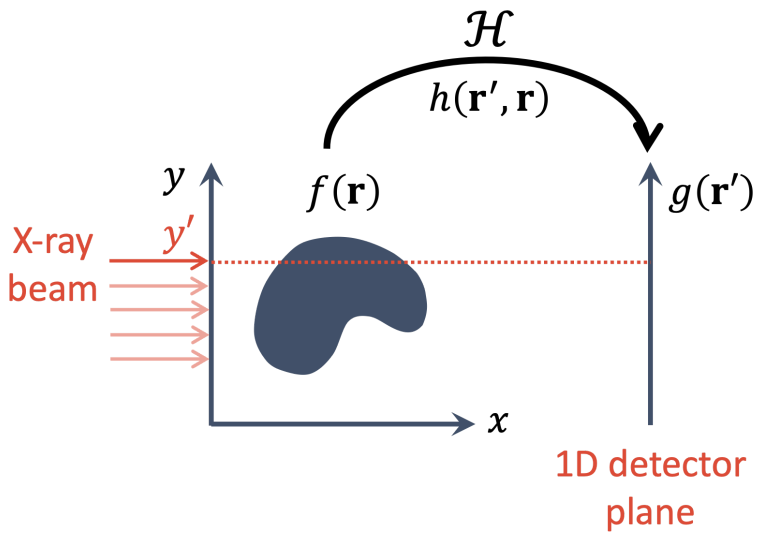
The X-ray propagation operator is characterized as $\mathcal{H} : \mathbb{L}_2(\mathbb{R}^2) \rightarrow \mathbb{L}_2(\mathbb{R}^1)$, where $\mathbf{r} = (x) \in \mathbb{R}^1$ represents a 2D sptial coordinate, and $\mathbf{r}' = (x', y') \in \mathbb{R}^2$ represents a 1D projection coordinate.

The forward projection of an object function $f(\mathbf{r})$ through an imaging system is given by:

$$g(\mathbf{r}') = \int d^2\mathbf{r}\, h(\mathbf{r}', \mathbf{r})\, f(\mathbf{r})$$

$$g(y') = \int\int dxdy\, \delta(y' - y) f(x, y)$$

$$= \int dx \int dy\, \delta(y' - y) f(x, y) = \int dx\, f(x, y')$$



The back-projection operator (adjoint of the operator $\mathcal{H}$) is:

$$[\mathcal{H}^\dagger g](\mathbf{r}) = \int d\mathbf{r}' h^*(\mathbf{r}, \mathbf{r}')\, g(\mathbf{r}')$$

$$[\mathcal{H}^\dagger g](x, y) = \int dy' \delta^*(y - y')\, g(y')$$

$$= \int dy' \delta(y - y')\, g(y') = \hat{f}\,(x, y)$$

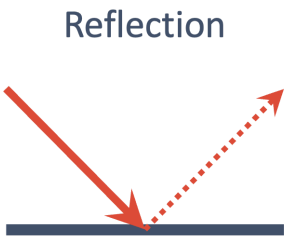
Behaviors of Traveling Waves

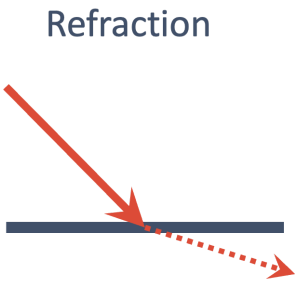
Regardless of obstacle size:

- **Absorption**: Light excites atoms, increasing vibrational energy, which is emitted as thermal radiation.

For obstacle size much larger than the optical wavelength ($\gg \lambda$):

- **Reflection**: Light bounces off the object surface.
- **Refraction**: Light changes direction when passing through a different medium due to a change in its progation speed.

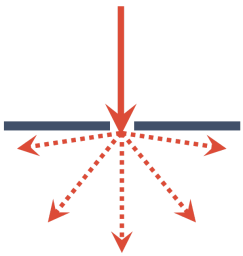




Obstacle size comparable to the wavelength ($\approx \lambda$):

- **Scattering**: Light deviates in multiple directions depending on object structure and wavelength.
- **Diffraction**: Light bends around an obstacle and can interfere constructively (larger amplitude) or destructively (smaller amplitude).





Wave Speed

The **speed of a wave** depends on the properties of the medium.

- **Light speed** in a medium is given by:

$$v = \frac{c}{n},$$

  where c is speed of light in a vacuum, and n is the refractive index of the medium.

- **Sound speed** in a medium is given by:

$$v = \sqrt{\frac{B}{\rho}},$$

where B is the bulk modulus of the medium (Pa=J/m^3 = (kg·m^2/s^2)/m^3), a measure of the medium's resistance to compression, and ρ is the density of the medium (kg/m^3).

Wave Intensity

The **intensity of a wave** is defined as:

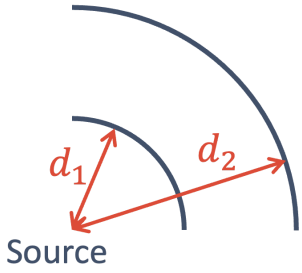
$$I = \frac{\text{Power}}{\text{Area}}$$

with units W/m^2 (J/(s·m^2)).

Here,

- $\text{Power} = (I)(\text{Area}) = \frac{E}{t}$: Power carried by the wave (W=J/s)
- E: Energy (J)

For a lossless system, power remains consistent over different distances: Power at distance $d_1 = \text{Power}$ at distance d_2, following the inverse-square law:



$$I_1(4\pi d_1^2) = I_2(4\pi d_2^2)$$

$$\frac{I_1}{I_2} = \frac{d_2^2}{d_1^2} \quad \text{Inverse-square law}$$

Beer-Lambert Law

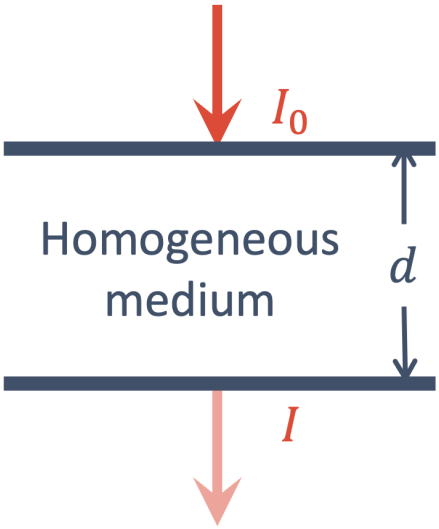
As waves propagate through a medium, attenuation (energy loss) occurs.

For a *homogeneous* medium, the **Beer-Lambert law** describes this attenuation as:

$$I = I_0 \exp(-\mu d),$$

where

- I_0: Incident light intensity (W/m^2=J/(s·m^2))
- d: Propagation distance (m)
- μ: Optical attenuation coefficient (1/m)



The effective optical attenuation coefficient is defined as:

$$\mu_{eff} = \sqrt{3\mu_a(\mu_a + \mu_s')} = \sqrt{3\mu_a[\mu_a + (1-g)\mu_s]},$$

where

- μ_a: Optical absorption coefficient (1/m)
- μ_s: Optical scattering coefficient (1/m)
- μ_s': Reduced optical scattering coefficient (1/m)
- g: Scattering anisotropy (dimensionless), a measure of the amount of forward direction retained after a single scattering event

Tomography

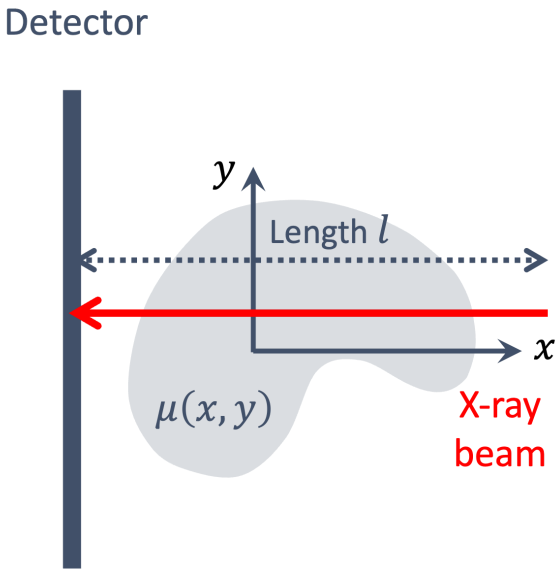
In X-ray imaging, an object f is represented by its distribution of optical attenuation coefficients $\mu(x, y)$

For a *homogeneous* medium, the **Beer-Lambert law** applies:

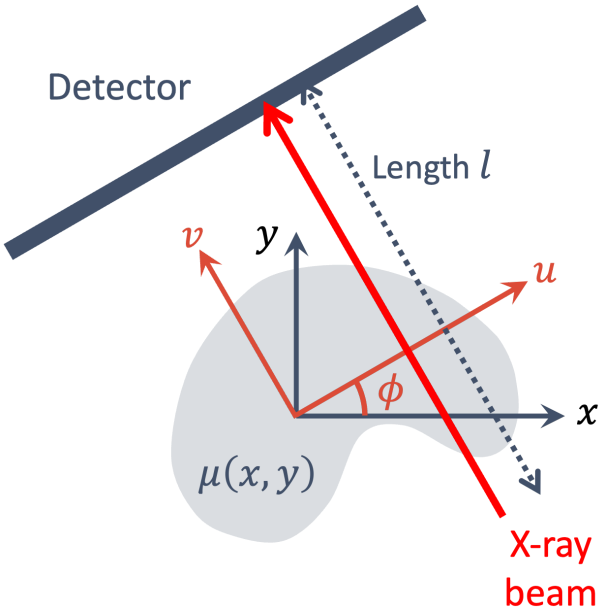
$$I = I_0 \exp(-\mu l)$$

For a *heterogeneous* medium, the **Beer-Lambert law** is path-dependent:

$$I = I_0 \exp(-\int_l dx\, \mu(x, y))$$

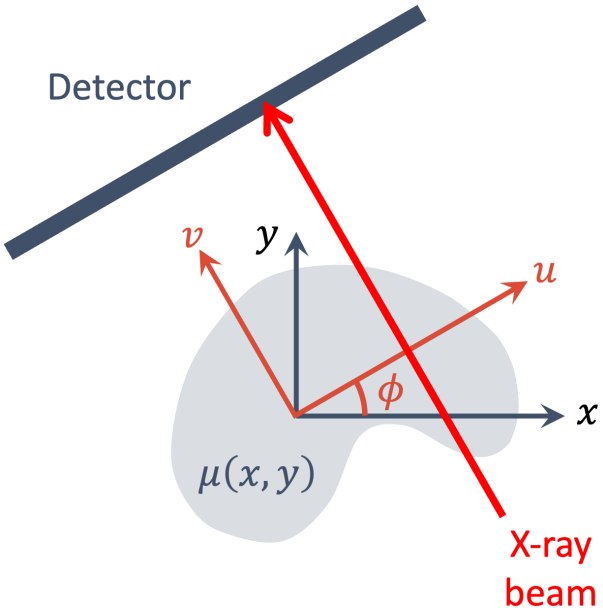


Tomography: By acquiring projection data from multiple angles ϕ around the object, an image can be reconstructed to visualize cross-sectional slices of the object.



Radon Transform

The intensity of light detected at a tomographic view ϕ is given by:



$$I(u,\phi) = I_0 \exp\Big(-\int_l dv\,\mu(x,y)\Big)$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$= I_0 \exp\Big(-\int_l dv\,\mu[u\cos(\phi) - v\sin(\phi), u\sin(\phi) + v\cos(\phi)]\Big)$$

To simplify, define the **logarithmic projection**:

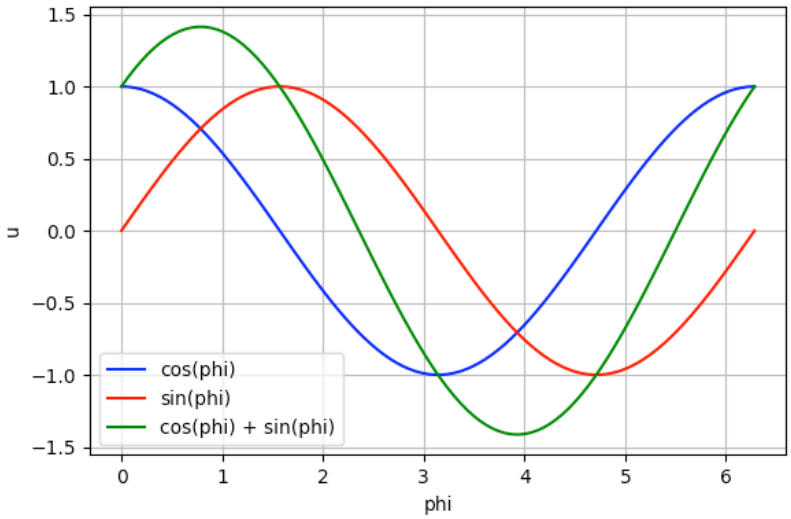
$$s(u,\phi) \equiv -\ln\frac{I}{I_0} = \int_l dv\,\mu(x,y)$$

$$= \int_l dv\,\mu[u\cos(\phi) - v\sin(\phi), u\sin(\phi) + v\cos(\phi)]$$

The **Radon transform** maps from the object to its sinogram, denoted as: $RT(f) = s(u,\phi)$, which corresponds to $\mathcal{H}f = g$.

Geometric Interpretation

For specific points:



$$(x,y) = (1,1),\ \text{then } u = \cos\phi + \sin\phi$$
$$(x,y) = (1,0),\ \text{then } u = \cos\phi$$
$$(x,y) = (0,1),\ \text{then } u = \sin\phi$$

At each angle ϕ, each point (x,y) contiributes to a corresponding projection coorindate u.

A **sinogram** is a collection of projections at different angles ϕ (stacked sine waves).

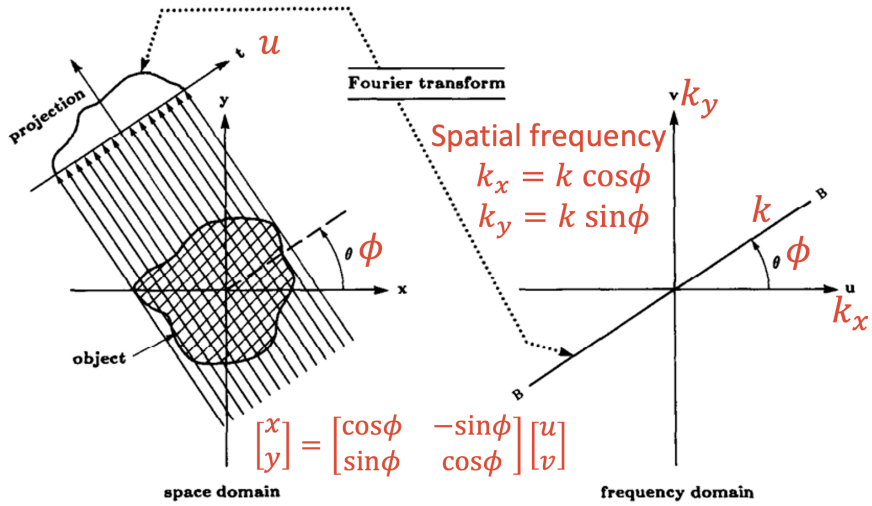
Central Slice Theorem (Fourier Slice Theorem)

The 2D Fourier transform of the object $f(x,y)$ is:

$$\mathcal{F}_{2D}[f(x,y)] = F(k_x,k_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dx\,dy\,f(x,y)\,e^{-i2\pi(k_x x + k_y y)}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dx\,dy\,f(x,y)\,e^{-i2\pi[(k\cos\phi)x + (k\sin\phi)y]}$$

Image from A. C. Kak and Malcolm Slaney, *Principles of Computerized Tomographic Imaging,* Society of Industrial and Applied Mathematics, 2001



$$\text{Jacobian determinant:} \quad dx\,dy = |\mathbf{J}|\,du\,dv$$

$$= \left| \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \right| du\,dv$$

$$= \left| \begin{bmatrix} \frac{\partial(x\cos\phi + y\sin\phi)}{\partial x} & \frac{\partial(x\cos\phi + y\sin\phi)}{\partial y} \\ \frac{\partial(-x\sin\phi + y\cos\phi)}{\partial x} & \frac{\partial(-x\sin\phi + y\cos\phi)}{\partial y} \end{bmatrix} \right| du\,dv$$

$$= \left| \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix} \right| du\,dv = du\,dv$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} du\,dv\, f(x,y)\, e^{-i2\pi[(k\cos\phi)(u\cos\phi - v\sin\phi) + (k\sin\phi)(u\sin\phi + v\cos\phi)]}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} du\,dv\, f(x,y)\, e^{-i2\pi[(ku\cos^2\phi - kv\cos\phi\sin\phi) + (ku\sin^2\phi + kv\cos\phi\sin\phi)]}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} du\,dv\, f(x,y)\, e^{-i2\pi(ku\cos^2\phi + ku\sin^2\phi)}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} du\,dv\, f(x,y)\, e^{-i2\pi(ku)}$$

$$= \int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} dv\, f(x,y)\, e^{-i2\pi(ku)} \quad \text{Sinogram: } s(u,\phi) = \int_{-\infty}^{\infty} dv\, \mu(x,y)$$

$$= \int_{-\infty}^{\infty} du\, s(u,\phi)\, e^{-i2\pi ku}$$

$$= S(k,\phi) = \mathcal{F}_{1D}[s(u,\phi)] \quad \text{1D Fourier transform of the sinogram}$$

X-ray Tomography

In practical X-ray imaging, we obtain a sinogram, but only a finite number of projections can be collected. The discrete sampling follows:

- u ➡️ $n\Delta u$, where Δu corresponds to the detector element size
- ϕ ➡️ $m\Delta\phi$, where $\Delta\phi$ corresponds to the tomographic view interval

Reconstruction Process

1. Compute the 1D Fourier transform of the sinogram:

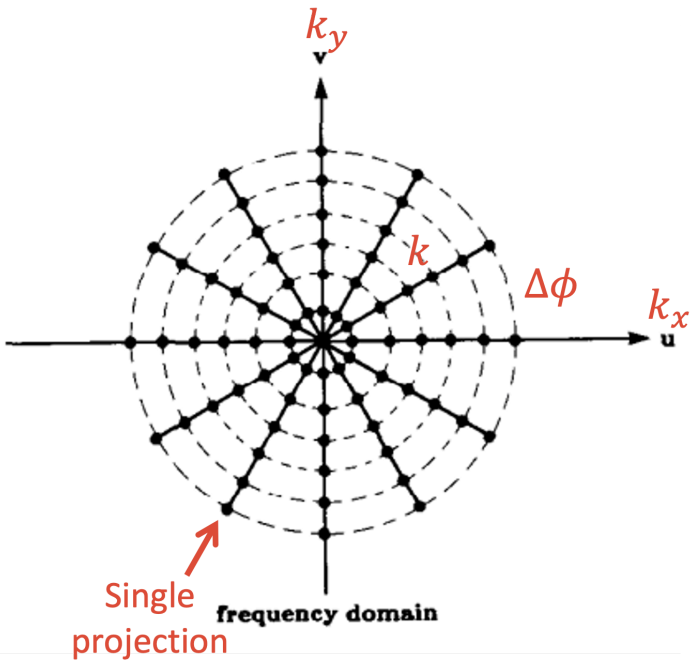
$$\mathcal{F}_{1D}[s(u,\phi)] = F(k_x, k_y)$$

2. Apply the 2D inverse Fourier transform to reconstruct the object:

$$\mathcal{F}_{2D}^{-1}[F(k_x, k_y)] = \hat{f}\,(x, y)$$

Filtered Backprojection (FBP)

A filter is applied to improve image quality, e.g., $S_{\text{filtered}}(k,\phi) = |k|S(k,\phi)$.



Images from A. C. Kak and Malcolm Slaney, *Principles of Computerized Tomographic Imaging,* Society of Industrial and Applied Mathematics, 2001



Ideal situation

What is actually measured

FBP applies a weighting in the frequency domain so that the data in (b) are an approximation to those in (a)

Challenges in Tomography

Several factors affect the quality and accuracy of tomographic reconstructions:

- Field-of-view: Limited scanning range restricts data acquisition.
- Undersampling: Sparse Δu and $\Delta \phi$ leads to artifacts.
- Scattering
- Partial volume effect: Discretization blurs object boundaries.
- Source (beam) shape: Wide beam angles introduce magnification.
- Source quality: Monochromatic sources improve contrast.
- Beam hardening: Low-energy photons get absorbed, altering intensity distribution in dense objects.

